

Laser-induced phase transitions of $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin films used in optical and electronic data storage and in thermal lithography

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Abstract: Amorphous thin films of $\text{Ge}_2\text{Sb}_2\text{Te}_5$, sputter-deposited on a ZnS-SiO_2 dielectric layer, are investigated for the purpose of understanding the structural phase-transitions that occur under the influence of tightly-focused laser beams. Selective chemical etching of recorded marks in conjunction with optical, atomic force, and electron microscopy as well as local electron diffraction analysis are used to discern the complex structural features created under a broad range of laser powers and pulse durations. Clarifying the nature of phase transitions associated with laser-recorded marks in chalcogenide $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin films provides useful information for reversible optical and electronic data storage, as well as for phase-change (thermal) lithography.

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1. Introduction

Phase-change materials are presently used as the recording layer in rewritable optical disks [1–3], as the storage cell in phase-change electronic memories [4–6], and as the inorganic resist layer in nano-scale thermal lithography [7–9]. The appeal of phase-change materials and their demonstrated capabilities are due in large measure to the ease and reversibility of the phase-transition between their amorphous and crystalline states [10,11]. What makes this

class of materials attractive for a diverse range of applications are the substantial differences that exist between the amorphous and crystalline states of many members of the class in their optical, mechanical, electronic, thermal, and chemical properties [12–15]. For example, the large difference in optical reflectivity between the amorphous and crystalline compounds of Ge, Sb, Te (perhaps doped or alloyed with Ag, In, etc.), are exploited in phase-change rewritable optical disks (CD, DVD, Blu-Ray) [16,17]. Differences in electrical conductivity of certain chalcogenide alloys and compounds provide the basis for their application in electronic phase-change random access memories (PCRAM) [18,19]. The fact that crystalline thin films of Ge, Sb, Te (with or without certain additives and dopants) etch rapidly in certain chemical environments, while the amorphous phase of the same material remains essentially intact, enables their application in phase-change lithography [20,21]. These examples suffice to explain the enormous current interest in phase-change materials, in their fabrication and characterization methods, and in their physical as well as chemical properties.

$\text{Ge}_2\text{Sb}_2\text{Te}_5$ is a representative chalcogenide compound that has been widely studied and used in optical and electronic phase-change memories. This material, which has a high degree of thermal, chemical, and mechanical stability, is also remarkable for its rapid transitions between crystalline and amorphous states [22,23]. In optical recording applications, where $\text{Ge}_2\text{Sb}_2\text{Te}_5$ is a viable candidate for the storage medium, the shape of the recorded mark is known to have a strong influence on the quality of the readout signal [24]. In near-field optical disk researches, the required writing and readout laser powers have been found to be outside the operation window of commercial systems [25–31]. Therefore, it is important to investigate the structural features of various types of recorded mark under a broad range of writing conditions, in order to gain a better understanding of the physical mechanisms involved in mark formation and stability [32–34].

This paper reports the results of our investigations into the structure of recorded marks on $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin films under a broad range of laser irradiation conditions. As a tool in these studies, we have found that the difference in the etch-rates of crystalline and amorphous $\text{Ge}_2\text{Sb}_2\text{Te}_5$ submerged in an NaOH solution provides a simple means of determining the structural features of recorded marks [35]. Optical and atomic force microscopy (AFM) [36–38] as well as scanning and transmission electron microscopy (SEM and TEM) have been used to confirm the detailed structure of each region for establishing a model for recorded marks.

2. Experimental

Using a conventional sputtering machine (Shibaura) with a working argon pressure of 5×10^{-1} pa, we deposited thin-film stacks with the structure of Substrate/ $\text{ZnS-SiO}_2/\text{Ge}_2\text{Sb}_2\text{Te}_5$ on fused-silica glass substrates. The $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST for short) material is a standard phase-change optical recording medium, while the ZnS-SiO_2 dielectric provides an appropriate subbing layer as well as the means for thermal control of the laser-marking and erasure processes on the GST material. Samples with crystalline-state $\text{Ge}_2\text{Sb}_2\text{Te}_5$ are subsequently obtained by annealing the sputter-deposited amorphous material in an oven at 300°C for 15 minutes.

An array of “marks” is written on the as-deposited $\text{Ge}_2\text{Sb}_2\text{Te}_5$ film using an optical pump-probe system (Static Media Tester, TOPTICA Co., Munich, Germany). This system uses a red laser beam (wavelength $\lambda = 658\text{nm}$), focused through a 0.65 NA objective. Laser power is varied from 1mW to 20 mW, while the laser pulse duration ranges from 100ns to 1500ns. The recorded marks are examined under an optical microscope, a transmission electron microscope (TEM, PHILIPS, Tecnai F30), and an atomic force microscope (AFM, Asylum Research). The recorded marks were further subjected to a chemical etching process in an NaOH solution with 1.0 wt% concentration and with different etching times. The NaOH solution is known to provide different etching rates for crystalline and amorphous states of the $\text{Ge}_2\text{Sb}_2\text{Te}_5$ material. The etched samples were re-examined for changes in the structure and morphology of the recorded marks under AFM and scanning electron microscope (SEM).

3. Result and discussion

Figure 1 shows the measured etch-rates of the as-deposited and annealed (i.e., crystalline) $\text{Ge}_2\text{Sb}_2\text{Te}_5$ films in 1.0 wt% NaOH solution. The sample structure used for this characterization was Glass/ ZnS-SiO_2 (130nm)/ $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (100nm). The etching time ranged from 30 minutes to 150 minutes, and the rotation rate of the magnetic stirring bar was 550 rpm. The solid rhombuses and triangles in Fig. 1 denote, respectively, the thickness of the as-deposited and crystalline samples after etching for the duration of time depicted on the horizontal axis. The chemical properties of the as-deposited state (essentially amorphous) are seen to differ substantially from those of poly-crystalline GST. Figure 1 shows that, for each sample, the relation between residual thickness and etching time is linear, with the slope being $7.2 \text{ \AA/min}$ for the crystalline state, and $2.3 \text{ \AA/min}$ for the as-deposited amorphous state. The etch-rate of crystalline GST is thus nearly three times that of the as-deposited amorphous film, with the result that, after 150 minutes of etching, the thickness difference between the two films is nearly 90nm (of which about 15nm is attributable to shrinkage upon crystallization, and the remaining 75nm is due to the difference of the etch-rates). This difference in etch-rate provides a mechanism for us to distinguish the two states of the GST material created in the process of laser-marking.

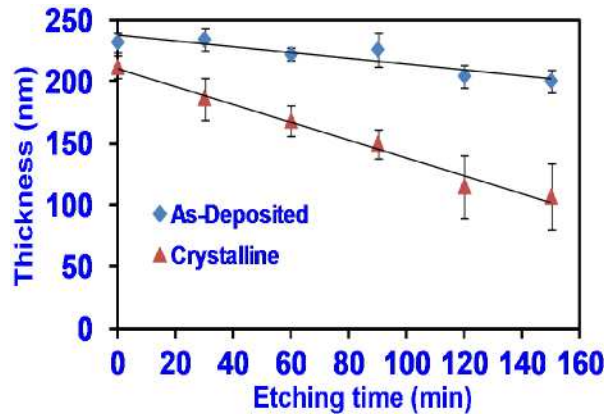


Fig. 1. Etching characteristics of sputter-deposited $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin films in 1.0 wt% NaOH solution under magnetic stirring at 550 rpm. The initial thickness of the as-deposited (amorphous) sample was 230nm, which is the sum of 130nm for the ZnS-SiO_2 dielectric layer and 100nm for the GST film. The annealed (poly-crystalline) sample was slightly thinner, at ~215nm, with its GST film shrunk by almost 15% upon crystallization. The etch-rate for each sample is seen to be constant over time, with the crystalline material being etched away at a rate of $\sim 7.2 \text{ \AA/min}$, more than three times that of the amorphous film at $\sim 2.3 \text{ \AA/min}$.

To investigate the formation mechanism of marks produced by laser irradiation of GST films, we recorded an array of marks on a Glass/ ZnS-SiO_2 (130nm)/as-deposited $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (50nm) sample using a range of laser powers (1-20mW) and pulse durations (100-1500ns). In Fig. 2(a), a reflection photo-micrograph, and Fig. 2(b), an AFM image, the separation between adjacent marks is about 3.0microns. Figure 2(c) is also an AFM image, but obtained

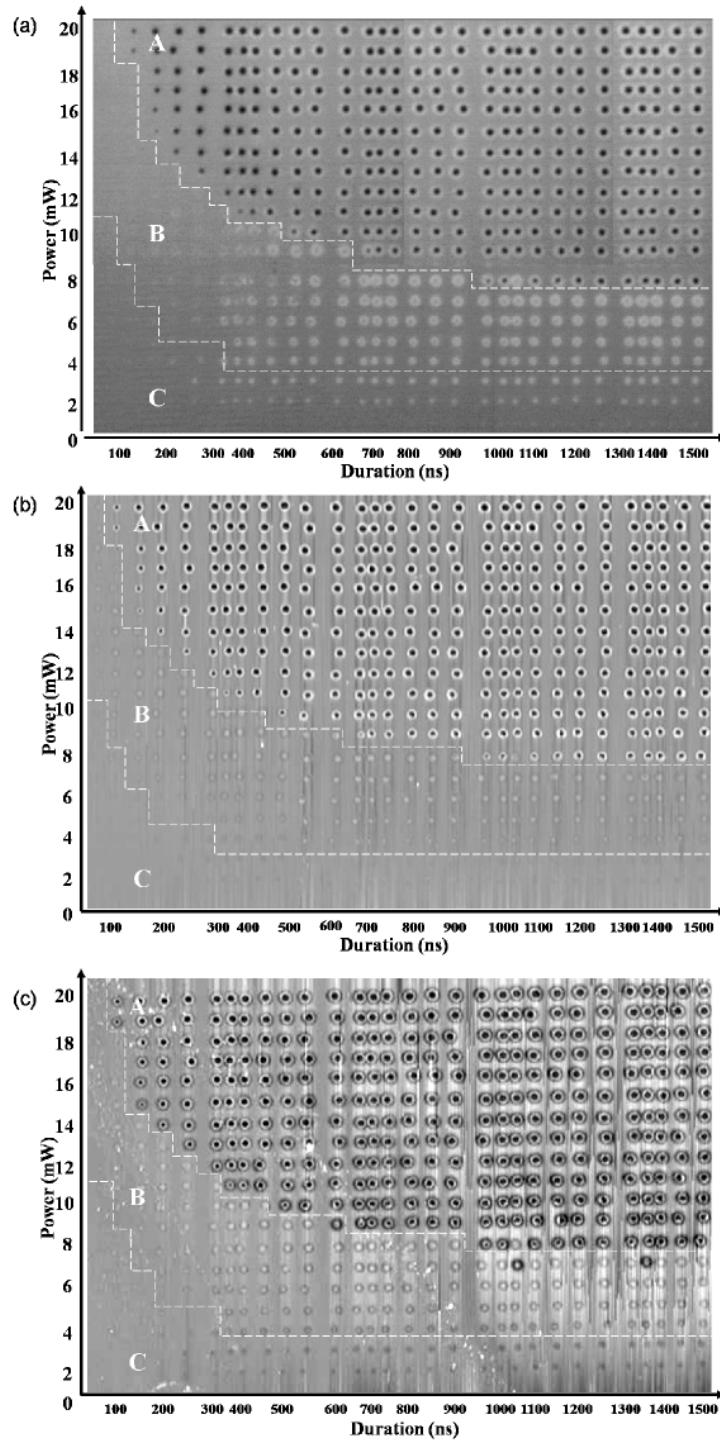


Fig. 2. Optical and atomic-force microscope images of marks recorded with different laser powers and different pulse durations on an as-deposited amorphous $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin film. For each mark, the pulse duration (nanoseconds) is indicated on the horizontal axis, while the laser power (milliwatts) appears on the vertical axis. (a) Optical micrograph and (b) atomic-force microscope (AFM) image of a $60 \times 90 \mu\text{m}^2$ area of the sample before etching. (c) AFM image of the same region of the sample following a 60 minute etch.

after etching the sample in a 1.0 wt% NaOH solution for 60 minutes. In each case the horizontal axis identifies the pulse duration, while the vertical axis gives the laser power used to record the corresponding mark.

The recorded marks shown in Fig. 2 can be divided into three categories, denoted by A, B, and C. Marks in category A, recorded at high laser power, show a hole in the middle, where the GST material has evaporated (or otherwise drifted away) upon melting. The bright ring surrounding the dark center of each mark consists, for the most part, of melt-quenched amorphous material, although the amorphous nature of this ring is not apparent from these pictures. Category B marks appear as light-gray spots on a dark grey background in both the optical micrograph of Fig. 2(a) and the AFM image of Fig. 2(b). As will be shown below, these marks are mostly crystalline GST with a melt-quenched amorphous region at their center, although the optical and atomic-force images of Figs. 2(a) and (2b) are incapable of resolving the detailed structure of these marks. Marks in category C – recorded either at low laser power with a long pulse, or at medium laser power with a short pulse – again appear in Fig. 2(a) as light-grey spots on a dark grey background, albeit smaller and fainter than those in category B. Upon further investigation, these category C marks will turn out to be poly-crystalline with no amorphous core, although the depth of the poly-crystalline region, depending on the laser pulse power and duration, could be less than the 50nm thickness of the GST film. The crystalline nature of category C marks may be inferred from their higher optical reflectance (relative to the background amorphous material) in the photo-micrograph of Fig. 2(a). The fact that these marks are invisible in the AFM image of Fig. 2(b) indicates that their geometrical depth, a consequence of shrinkage upon crystallization [32], is rather shallow.

Applying a similar logic to the AFM image of category B marks, we note that in Fig. 2(b) these marks appear brighter than their background, indicating the presence of a bump (as opposed to a pit) at the core of these recorded marks. The bumps are smaller than the corresponding optical image of category B marks in Fig. 2(a), indicating that only the core of the recorded mark has risen above the surface of the GST layer. These cores being amorphous in their atomic structure (as will be shown below), one expects their optical image in Fig. 2(a) to have a small dark-grey spot at the center of the bright-grey image of the poly-crystalline ring that surrounds them. Indeed most of the category B marks in Fig. 2(a) show hints of this amorphous core, although the core is apparently too small to be fully resolved by our optical microscope.

Next we compare the images of the sample before etching with the AFM image obtained after a 60-minute etch, shown in Fig. 2(c). By and large, the etching process removes the crystalline regions of the recorded marks, leaving behind the amorphous regions essentially intact. For category A marks in Fig. 2(c), we notice that etching has created a thin, dark ring around the bright rings that were previously suspected of being amorphous in nature. The exterior regions of category A marks, surrounding the bright rings of Figs. 2(b) and (2c), are thus revealed to be poly-crystalline. This is reasonable, considering that the focused laser beam illuminating these exterior regions is too weak to melt the GST, yet strong enough to anneal the material into its poly-crystalline state.

Category B marks also develop, upon etching, a dark ring around their central bright core, as seen in the AFM image of Fig. 2(c). The distinguishing feature of these marks is thus revealed as being a shallow crystalline ring surrounding a central core of melt-quenched amorphous GST, which core is in the form of a bump that rises somewhat above the film surface. Finally, marks in category C are completely etched away in the AFM image of Fig. 2(c), as they consist of nothing but crystalline grains formed in consequence of annealing under the low-power focused laser beam.

Scanning electron micrographs of the recorded marks are shown in Fig. 3. These SEM images generally confirm the conclusions reached by analyzing the optical and atomic-force images in the preceding paragraph, but they do not contribute much additional information. What is noteworthy perhaps is that, due to the greater resolution of the SEM images, the void at the center of category A marks is distinctly visible in Figs. 3(b) and 3(e), where a small

deposit of left-over debris from the molten pool can be recognized at the bottom of the void. Upon etching, the left-over debris disappears, along with the exterior portion of the ring that surrounds the central void; see Fig. 3(f).

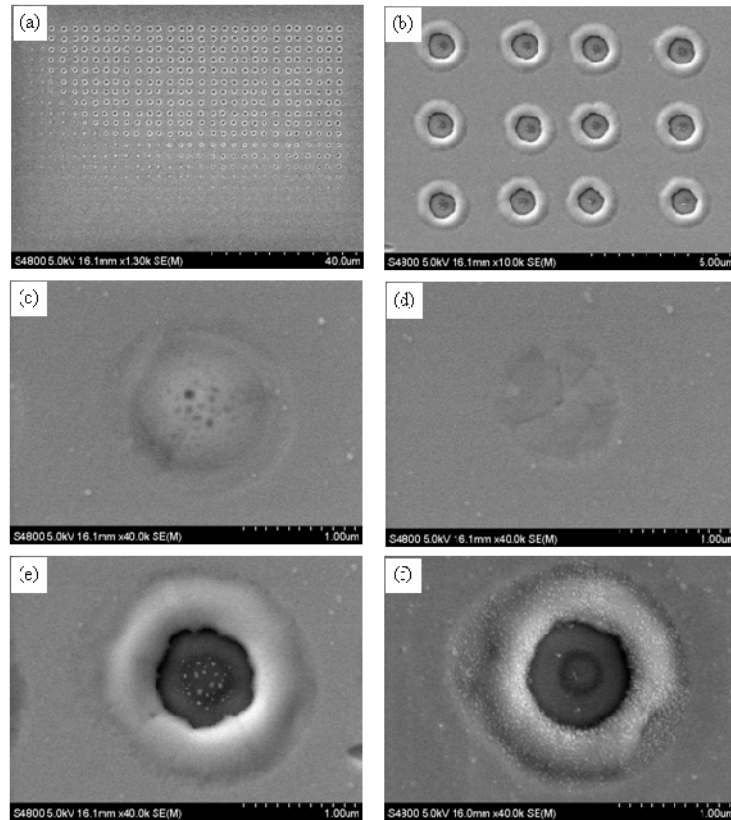


Fig. 3. Scanning electron micrographs of recorded marks in a 50nm-thick $\text{Ge}_2\text{Sb}_2\text{Te}_5$ film on a 130nm-thick ZnS-SiO_2 dielectric layer deposited on a glass substrate. The marks imaged in (a) were recorded under the same conditions as those in Fig. 2. (b) Close-up image of several category A marks shows the void at the center of the mark and a ring surrounding the void. (c) SEM image of a category B mark shows a few dark spots near the center of what appears to be a bulged-up region surrounded by a slightly depressed ring. (d) Image of a category C mark is unremarkable except for the appearance of a slightly depressed region in the area previously illuminated by the focused laser beam. (e) Magnified view of a category A mark, written with a 16mW and 1100ns laser pulse, showing a few bright spots at the bottom of the central void, in what appears to be a remnant of the molten pool formed at the center of the mark during laser-writing. (f) SEM image of a category A mark after etching. The exterior boundary of the ring as well as the remnants of the molten pool at the bottom of the void have been etched away. The nature of the small white dots sprinkled all over the ring is not presently understood.

To confirm the initial results obtained from the above “chemical identification” of amorphous and crystalline regions by means of their differing etch-rates, and also to investigate the mark-formation process in greater detail, we conducted transmission electron microscopy (TEM) combined with local electron diffraction analysis on our GST samples. For TEM observations, we sputter-deposited our bilayer stack, ZnS-SiO_2 (130nm)/ $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (50nm), on a copper grid coated with a thin carbon support layer. We found the structure of the recorded marks on this sample and their dependence on laser power and pulse duration to be similar, although by no means identical, to those obtained with samples fabricated on glass substrates. In general, the carbon support layer being extremely thin (on the order of a few ten nanometers), the thermal energy deposited by the laser pulse stays for a longer time within the GST layer, and also diffuses laterally rather than dissipating through the substrate. The net

result is that, marks recorded in the GST film atop the carbon layer, are generally larger than those recorded under the same conditions on a glass substrate. Moreover, since the cooling process on the carbon support layer is rather slow, the crystalline regions in samples prepared for TEM observations tend to grow at the expense of melt-quenched amorphous regions, which shrink in size relative to those obtained on glass substrates. Nevertheless, marks recorded on the GST film atop the carbon support layer could still be divided into the three main categories described in Fig. 2.

Figure 4 shows the TEM image as well as local electron diffraction patterns obtained for a mark that was recorded with a 20mW and 700ns laser pulse. [Note that this mark is nearly 5 μ m in diameter, compared to only 2 μ m for the mark in Fig. 5(e), which was recorded under similar conditions, but on a glass substrate.] In general, due to differences in the electron density, crystal orientation, and composition of the material, the crystalline regions of the sample appear darker in the TEM image than do the amorphous regions. The region marked with the letter “d” in the TEM image of Fig. 4(a) may thus be identified as a crystalline region with large crystal grains. The electron diffraction patterns associated with different regions of the recorded mark also exhibit substantially different characteristics, indicating the existence of different phases of the GST material within a given recorded mark. The diffuse halo of the electron diffraction pattern in Fig. 4(b) reveals the sample background as being in the as-deposited amorphous state. Figure 4(c) shows a diffuse halo pattern sprinkled with several spots from randomly oriented crystalline grains; the outer regions of the mark are thus seen to have small crystallites distributed within an amorphous network. It is also entirely possible that, in region c as well as in the outer zone of region d, a crystalline GST layer under lays an amorphous surface layer, for the simple reason that the electron beam, as it traverses the GST film along the surface normal, must interact with everything in its path.

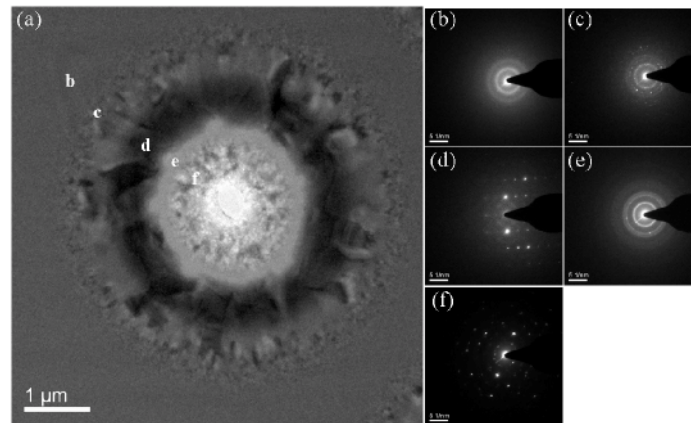


Fig. 4. (a) Transmission electron micrograph, and (b-f) in situ electron diffraction patterns obtained from various regions of a mark recorded on an as-deposited Ge₂Sb₂Te₅ thin film using a 20mW and 700ns laser pulse. The GST film and its ZnS-SiO₂ subbing layer were deposited on a copper grid coated with an extremely thin carbon diaphragm. The recorded mark, although created under conditions that were similar to those used to write the mark of Fig. 3(e), is substantially larger than the latter. Moreover, the relative diameters of the amorphous and crystalline rings are different in the two cases in consequence of the differing heating and cooling cycles each mark has been subjected to. The regions identified with letters “b” through “f” in the TEM image correspond to the diffraction patterns shown in frames (b)-(f), respectively.

The regular diffraction spots from a single-crystalline area shown in Fig. 4(d) confirm that the dark ring seen in the TEM image of Fig. 4(a) is indeed poly-crystalline. The electron diffraction pattern obtained from region “e” [inside the dark ring of Fig. 4(a)] and displayed in Fig. 4(e) is similar to that obtained from the as-deposited background region, albeit with sharper, perhaps slightly less diffuse, rings. The atomic structure of region “e” is therefore identified as melt-quenched amorphous, which is what we associated earlier with the bright

doughnuts seen in category A marks of the AFM image in Fig. 2(b). The amorphous ring labeled as e in Fig. 4(a) is narrower than the doughnuts of Fig. 2(b), probably because of the slower cool-down of the molten GST pool atop the carbon support layer.

Finally, the diffraction pattern of Fig. 4(f) represents the inner ring of the mark, which surrounds the central void and apparently consists of small, randomly-oriented crystallites. The electron diffraction patterns in Fig. 4, being mixtures from several grains encountered through the film thickness, are not clear enough for us to determine the crystalline structure in regions c, d, and f. Based on the reported lattice structure of GST films annealed at different temperatures, however, we conjecture that the crystallites of region b, which do not reach very high temperatures during laser-writing, are face-centered cubic (fcc), whereas those in regions d and f, having been formed at much higher temperatures, are probably hexagonal.

For a category B mark, recorded with a 6mW and 700ns laser pulse, Fig. 5(a) shows the TEM image and, in the inset, the electron diffraction pattern obtained from the central region of the mark. The mark is clearly seen to be melt-quenched amorphous at the center and crystalline at the periphery, as argued in our earlier discussion in conjunction with Fig. 2. Note that the melt-quenched amorphous GST appears brighter in this TEM image than the as-deposited amorphous area of the background; clearly the amorphous state is not unique and, depending on the route taken to arrive at a particular amorphous state, one should expect to see different physical (and perhaps chemical) properties. Figure 5(b) shows the TEM image and, in the inset, the electron diffraction pattern from the central region, of a category C mark recorded with a 2mW and 700ns laser pulse. The recorded mark in this case is polycrystalline, showing relatively large crystallites of differing orientations, probably sitting atop an amorphous region of the GST film below the surface, which, due to the relatively low laser power used in this case, has not been sufficiently crystallized.

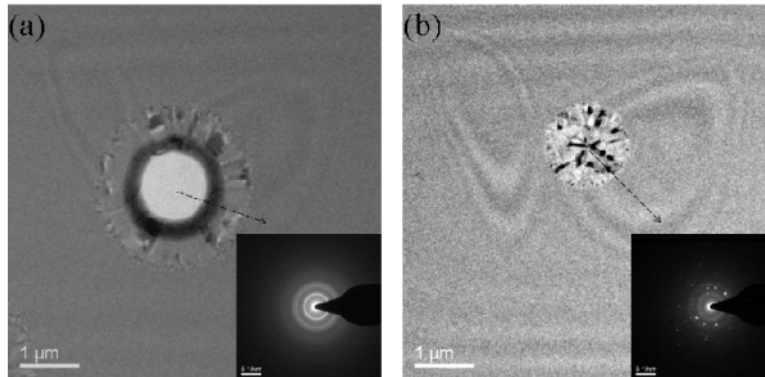


Fig. 5. Transmission electron micrographs and the corresponding electron diffraction patterns (inset) from the central region of two recorded marks on an as-deposited amorphous $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin film. Both marks were recorded with a 700ns laser pulse; the laser power was 6mW in (a) and 2mW in (b). The mark recorded at low power is polycrystalline, with black and white regions in the TEM image indicating the various orientations of the crystalline grains. In contrast, the mark recorded at high power has gone through melting and rapid quenching at its center, where the GST material is once again amorphous, albeit with a random atomic network that is somewhat different from that of the as-deposited material. The dark ring surrounding the bright central part of the TEM image in (a) is in the crystalline state.

We experimented with different rotational speeds of the magnetic stir bar during the etching process. Figure 6 shows the AFM image of recorded marks after 60 minutes of etching in a 1.0 wt% NaOH solution, with the stirring bar rotating at 870 rpm. As before, the sample structure was Glass/ZnS-SiO₂ (130nm)/as-deposited $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (50nm). For the relatively small region of the sample shown in Fig. 6, the range of the pulse duration (horizontal axis) was 1100-1300 ns, while the laser power (vertical axis) varied from 10mW to 14mW. These are all category A marks, which have a hole at the center (created during laser writing), a melt-quenched amorphous ring with the same height as the background (i.e.,

the as-deposited amorphous material of the GST film), and an exterior crystalline ring that has been etched away and, therefore, appears as a void (dark ring) in the AFM image of Fig. 6. Note that a few of the doughnut-shaped amorphous rings have been detached from the ZnS-SiO₂ subbing layer and floated a small distance away. It appears that the region under the amorphous ring may have been crystalline, which, upon severe etching and complete removal, has caused the detachment of the amorphous ring. Although other plausible explanations for the detachment exist, this could yet be an indication that, beneath the surface, the recorded mark's structure may differ from what one observes on the surface.

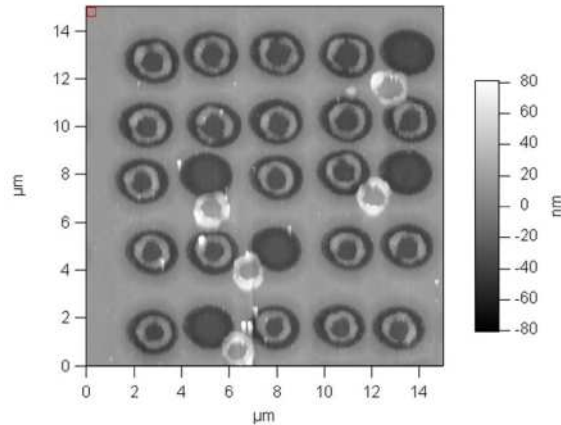


Fig. 6. Atomic force micrograph of a $15 \times 15 \mu\text{m}^2$ area of an as-deposited $\text{Ge}_2\text{Sb}_2\text{Te}_5$ sample after marks recorded with a number of laser pulses (10-14 mW, 1100-1300 ns) were subjected to a 60-minute etch process under a magnetic stirring rate of 870 rpm. The grayish, doughnut-shaped region within each mark is melt-quenched amorphous material that has survived the etching process. The dark region inside each mark is a hole created in the GST layer during laser writing. The dark rings that surround each doughnut are also voids, but these are created by the etching away of crystallized regions. A few of the (amorphous) doughnuts are seen to have been completely detached from the substrate and floated a little distance away from their original locations within the recorded marks. The voids are around 50nm deep, which is the thickness of the GST film. Also, the white doughnuts that have left their original locations – and are now sitting atop the as-deposited regions of the sample – have a height of 50nm, which is consistent with the original thickness of the GST layer.

Based on the above experimental observations, we believe the following model could describe the structure of laser-written marks in GST films. Figures (7a)-(7c) show cross-sectional diagrams of the three types of recorded mark. In Fig. 7(a), a hollow core and a melt-quenched amorphous ring supported and surrounded by a poly-crystalline ring is representative of marks recorded with high-energy laser pulses. Figure 7(b) shows that recording with laser pulses of moderate energy could result in a melt-quenched amorphous core region surrounded by a bowl-shaped crystalline boundary. Figure 7(c) pertains to the case of recording with low-energy laser pulses, where the marks are generally in the form of poly-crystalline pancakes. The schematic diagrams in the bottom row of Fig. 7 show what we believe to be the resulting structure upon etching the recorded marks depicted in the corresponding frame at the top of the figure.

4. Concluding remarks

We have studied the structure of phase-change marks recorded in sputter-deposited thin films of amorphous $\text{Ge}_2\text{Sb}_2\text{Te}_5$ under various laser powers and pulse durations. We observed the formation of ring-shaped domains of both amorphous and crystalline structure within the same mark. Optical, atomic-force, scanning electron, and transmission electron microscopy of the recorded marks, both before and after etching of the crystalline regions in an NaOH solution, provided insights into the structural features and complex processes involved in laser-induced mark formation in phase-change recording materials.

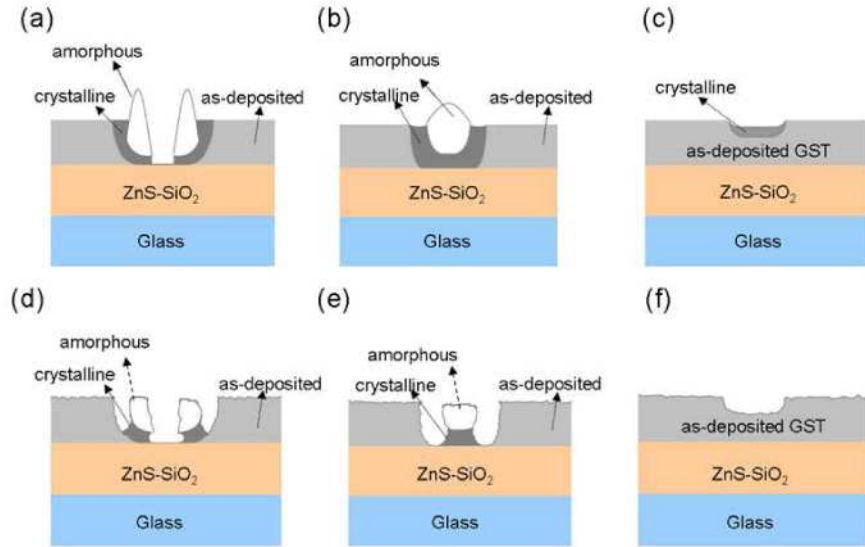


Fig. 7. Graphical description of the structure of various recorded marks before etching (top row) and after etching (bottom row). (a) At high levels of laser power and long pulse durations, the center of the recorded mark evaporates, the interior ring of molten material rapidly quenches into the amorphous state, while the exterior ring cools slowly to form a crystalline region. (b) At moderate laser powers and pulse durations, there is no central hole, and the consolidated amorphous region at the center is surrounded by crystalline material. (c) At low laser powers, the material never melts and, therefore, there is no quenching and re-amorphization. Only a shallow crystalline mark is recorded, whose depth and diameter both increase with an increasing laser power and/or pulse duration. Upon etching, crystalline regions of the recorded marks that are exposed to the etchant are slowly eroded, while the amorphous regions by and large survive the etching process. If etching continues for a longer time, the crystallites underlying the amorphous regions erode even further, causing the amorphous “ring” or “ball” to detach from the substrate.

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